

Part A	Basis:				moles in solution
	17.27615 kg of solution -->	10 kg water -->	0.010384 m ³ water ==>		575.3832 mol water
	0.6 wt% water	6.9 kg ethanol	0.008754 m ³ ethanol		149.9991 mol ethanol
	0.4 wt% ethanol		SUM 0.019139 m³ solution		

**Assuming perfectly additive volumes

Density

789.4 kg/m³ ethanol @ 20C

998.19 kg/m³ water @ 20C

MW

46.07 g/mol ethanol

18.01528 g/mol water

Part B	Basis:				moles in solution
	17.27615 kg of solution -->	10 kg water -->	0.018475 m ³ solution ==>		575.3832 mol water
	0.6 wt% water	6.9 kg ethanol			149.9991 mol ethanol
	0.4 wt% ethanol				

Specific Gravity

0.93518 (given for the mixture @ 20C)

Percent Error:

3.594240843 %

Density

999.95 kg/m³ water @ 4C (reference)

935.1332 kg/m³ mixture @ 20C (from S.G.)

MW

46.07 g/mol ethanol

18.01528 g/mol water